

JAVA SDE-2 INTERVIEW PREPARATION SERIES

SYSTEM DESIGN AND BACKEND - BOOK 09 OF 14 - STUDY STEP 19J

Apache Kafka and Spring Kafka

Logs, Partitions, Consumer Groups, Delivery Semantics, and Java Integration

BY

Vinay Reddy Kalluri

One coherent roadmap from Kafka fundamentals to Spring Kafka configuration, consumers, producers, retries, tests, and operations.

KAFKA | SPRING KAFKA | PARTITIONS | OFFSETS | DELIVERY

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

Series edition - Release 2026-07-29

Open educational interview-preparation edition

Start Here - Your Route Through This Volume

60-second entry rule: If two or more recognition signals below expose a gap, begin with Chapter 1. If the prerequisites are weak, step back to SD 08 – Redis. If you can already explain every outcome without notes, jump to Part II and prove it through the practice ladder.

Keep Apache Kafka semantics and Spring Kafka integration together while teaching the platform model before framework abstractions.

Readiness map

Decision	Evidence
START HERE	Services communicate asynchronously; Ordering, replay, or consumer scaling matters; Retries and delivery semantics can duplicate effects.
PREPARE FIRST	Spring Boot Book 05; Database transaction basics.
MOVE ON WHEN	Understand the planned Kafka-to-Spring-Kafka sequence; Connect partitions and offsets to application correctness; Know the production and interview depth required.

Choose the route that fits today

- 1 **Learning:** read in order, type the representative Java examples, and complete each dry run.
- 2 **Revision or rehearsal:** start at the first decision map, invariant, or failure mode you cannot explain; if none expose a gap, go directly to Part II and prove readiness without notes.

System Design Segment Position

SD 08 - Redis	YOU ARE HERE SD 09 - Apache Kafka and Spring Kafka	SD 10 - Spring Ecosystem Extensions
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Part I - Core Learning**SDE-2 CORE CHAPTER**

Chapter 1 - Apache Kafka and Spring Kafka - Planned Learning Roadmap

Publication status: roadmap edition. Runnable brokers, producer and consumer labs, rebalance drills, and delivery-semantics exercises will follow.

Apache Kafka is a distributed log. Spring Kafka integrates Kafka clients with Spring configuration, listener containers, transactions, retries, and testing support. The book will teach Kafka contracts first and Spring integration second.

Planned sequence

- 1 Topics, partitions, records, offsets, retention, compaction, and ordering scope.
- 2 Brokers, leaders, replicas, in-sync replicas, acknowledgments, and durability.
- 3 Producers, keys, partitioning, batching, compression, retries, and idempotence.
- 4 Consumers, groups, assignments, polling, commits, lag, and rebalances.
- 5 At-most-once, at-least-once, transactional processing, and practical deduplication.
- 6 Schema contracts, compatibility, poison records, retry topics, and dead-letter handling.
- 7 Spring Kafka factories, templates, listeners, error handlers, transactions, and tests.
- 8 Capacity, partition counts, hot keys, observability, upgrades, and incident recovery.

Interview focus

Readers will learn to define the ordering key, choose acknowledgment and commit boundaries, explain replay, handle duplicates, diagnose lag, and defend a retry and dead-letter policy. Framework annotations will always be tied back to native client behavior.

Completion gate

A reader is ready to design a Kafka workflow when they can state partitioning, delivery, replay, schema, failure, and observability contracts and can explain exactly where Spring Kafka starts and where Kafka client semantics still control correctness.

Part II - Practice and Handoff

Practice Ladder and Next Steps

TRY IT | Practice ladder

- Foundation: log, partitions, producers, and consumers
- Interview Core: groups, offsets, delivery, and Spring integration
- SDE-2 Follow-up: retries, transactions, observability, and incidents

For every coding problem, state the input contract, select the numeric and collection types deliberately, write the invariant before the loop or recursion, test the smallest and largest valid inputs, and close with time and auxiliary-space complexity.

Navigation

[Previous book: SD 08 - Redis for Java Backend Interviews](#)

[Series index](#)

[Next book: SD 10 - Spring Ecosystem Extensions](#)

TRY IT | Completion check

- ☐ I can recognize the core patterns without being told the data structure or algorithm.
- ☐ I can implement the representative Java templates from memory.
- ☐ I can explain why each implementation is correct.
- ☐ I can test boundaries, invalid inputs, overflow, and adversarial shapes.
- ☐ I can answer at least one SDE-2 follow-up involving trade-offs or production constraints.

Series Roadmap

Choose Java, DSA, or System Design and Backend first, then follow the books inside that segment in order. Stable study-step codes remain on filenames and links; the clearer segment book codes appear on the website and every PDF cover.

SEGMENT	BOOK	FOCUSED LEARNING PATH
Java	JAVA 01	Java Foundations for Problem Solving
Java	JAVA 02	Git and GitHub for Java Engineers
Java	JAVA 03	Maven and Gradle for Java Engineers
Java	JAVA 04	Advanced Java B: Language, OOP, and Modern Java
Java	JAVA 05	Advanced Java C: Collections, Streams, and I/O
Java	JAVA 06	Advanced Java A: JVM and Execution
Java	JAVA 07	Advanced Java D: Concurrency and the Memory Model
Java	JAVA 08	Advanced Java E: Performance, Diagnostics, and GC Incidents
Java	JAVA 09	Advanced Java G: Question Bank, Study Plan, and Reference
DSA	DSA 01	Time and Space Complexity for Java Interviews
DSA	DSA 02	Number Systems and Math Foundations for DSA Interviews
DSA	DSA 03	Number Systems Interview Patterns and Rapid Revision
DSA	DSA 04	Bit Manipulation in Java
DSA	DSA 05	Loop Mastery, Patterns, and Index Calculations
DSA	DSA 06	Arrays and Array Problem-Solving Patterns
DSA	DSA 07	Strings and String Problem-Solving Patterns
DSA	DSA 08	Hashing: Maps, Sets, Frequency, and Prefix State
DSA	DSA 09	Recursion and Backtracking in Java
DSA	DSA 10	Linked Lists: Pointer Reasoning and Mutation
DSA	DSA 11	Stacks, Queues, Deques, and Monotonic Patterns
DSA	DSA 12	Binary Search: Bounds, Answers, and Invariants
DSA	DSA 13	Trees, Binary Search Trees, and Tries
DSA	DSA 14	Heaps, Priority Queues, Selection, and Top-K
DSA	DSA 15	Graphs: Traversal, Ordering, Paths, and Union-Find
DSA	DSA 16	Greedy Algorithms: Recognition, Proofs, and Scheduling
DSA	DSA 17	Dynamic Programming: State, Transitions, and Optimization
System Design	SD 01	Advanced Java F: Design, Backend, Testing, and Security
System Design	SD 02	MySQL for Java Backend Interviews
System Design	SD 03	Hibernate and JPA for Java Backend Interviews
System Design	SD 04	Spring Framework for Java Engineers
System Design	SD 05	Spring Boot for Java Backend Engineers

SEGMENT	BOOK	FOCUSED LEARNING PATH
System Design	SD 06	Spring Data for Java Backend Engineers
System Design	SD 07	MongoDB for Java Backend Interviews
System Design	SD 08	Redis for Java Backend Interviews
System Design	SD 09	Apache Kafka and Spring Kafka
System Design	SD 10	Spring Ecosystem Extensions
System Design	SD 11	Spring AI for Java Engineers
System Design	SD 12	Advanced Java H: Spring Boot and REST APIs
System Design	SD 13	Advanced Java I: Persistence, SQL, JPA, and Caching
System Design	SD 14	Advanced Java J: Distributed Systems and System Design

Roadmap editions identify planned books whose chapter sets will be expanded one at a time. Existing deep-dive books remain available later in each segment, so no published material is displaced.

About the Author

Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer, the creator and founding author of the Java SDE-2 Interview Preparation Series, and its Editor-in-Chief and Chief Auditor. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

Editorial leadership

As Editor-in-Chief, Vinay owns the learning sequence, scope, voice, and publication decisions. As Chief Auditor, he owns the Java accuracy standard, validation evidence, attribution review, and release-readiness gate. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

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Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: System Design and Backend - Book 09 of 14 - Study Step 19J. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

Source provenance: curated from the independent Java SDE-2 master guide plus focused original material. Original master chapter numbers are labeled explicitly when retained in cross-references.